Modeling metaphor evolution on cancer online narratives

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Metaphors are linguistic devices that allow speakers to explain abstract experiences, such as being ill with cancer, in terms of more familiar and tangible domains, for example through war metaphors (*I am fighting cancer*) (Lakoff & Johnson, 1980). Given their pervasiveness (Casarett et al., 2010) and communicative importance in medical contexts (Semino et al., 2017), metaphors have been widely studied in oncological discourse (Liu et al., 2024). However, most existing research treats metaphors as relatively static linguistic phenomena, overlooking how they may emerge, spread, and evolve within communities over time.

Cancer, by contrast, is a prolonged and emotionally dynamic process characterized by successive phases: diagnosis, treatment, remission, relapse, and end-of-life; each involving shifting emotions, goals, and coping strategies (Sañudo et al., 2025). The language patients use to describe these experiences is therefore likely to evolve accordingly. Online cancer forums provide a unique naturalistic setting for observing real-time linguistic change, where metaphors can be traced as they are adapted, reinterpreted, and socially negotiated through interaction.

Building on dynamic approaches to metaphor grounded in complex systems theory (Cameron et al., 2009), this Stage 1 Registered Report conceptualizes metaphor use as a form of micro-scale language change driven by social interaction. We hypothesize that metaphorical expressions undergo processes analogous to linguistic evolution: variation through creative innovation, selection through social uptake or rejection, and retention through repeated use and conventionalization within communities.

The analysis draws on interaction data from the Asociación Española Contra el Cáncer (AECC) online forums. The corpus comprises posts authored by patients, relatives, and caregivers between 2010 and 2024, providing a diachronic window of fourteen years of naturally occurring health communication. All posts were publicly available and pseudonymized, containing no personal or demographic information. Given the AECC's national reach, the forum includes contributors from diverse socioeconomic, educational, and regional backgrounds.

The dataset consists of 7,964 threads ($\approx$ 5.6 million tokens), distributed across

three thematic categories: breast cancer, prostate cancer, and personal testimonies. Posts are semi-formal, asynchronous, and oriented toward emotional support and experience sharing. Each thread includes one main post followed by multiple responses, forming interactional chains in which linguistic adaptation can be observed. The corpus comprises interactions among approximately 3,000 users, some of whom have participated in over 1,000 posts across extended time spans, allowing us to trace longitudinal patterns in individual language use. Figure 1 shows a sample of the interaction network based on the 100 most interactive posts.

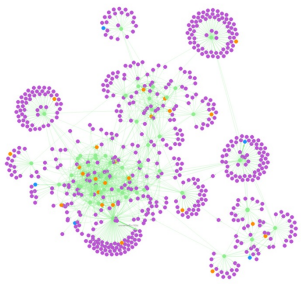

Figure 1. Network of main authors (orange) and commenters (blue) with top 100 posts with mosts interactions.

Methodologically, the project integrates corpus linguistics, computational modeling, and network analysis. Metaphor identification will rely on Contrast-WSD (Elzohbi & Zhao, 2024), trained on a manually annotated subset of over 80,000 tokens from the AECC forum. Annotation of metaphorical target domains allows us to distinguish well-established frames (e.g., warfare, journey) from less conventional domains (e.g., construction, education), which we treat as potential sites of metaphorical innovation. To model linguistic change, we construct temporal interaction graphs representing users, threads, and metaphor usage across successive time windows. Community detection and network measures (e.g., connectivity, assortativity, and centrality) are used to identify clusters of users, influential speakers, and pathways of metaphor diffusion. By comparing graph configurations across years and across phases of illness discourse, we quantify how metaphorical patterns reconfigure over time.

Preliminary analyses (Figure 2) illustrate how the centrality of different metaphor domains shifts diachronically. Journey metaphors remain consistently central, in line with previous literature (Liu et al., 2024). Violence metaphors, while initially central as reported in prior studies, appear to decrease in prominence over time. This decline may reflect broader sociocultural critiques of militaristic framing in illness discourse, potentially creating space for alternative

metaphorical constructions such as education and construction.

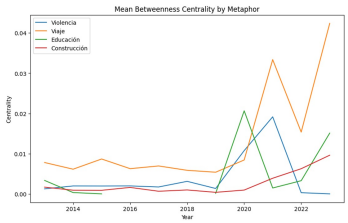

Figure 2. Mean Centrality of violence (blue), journey (orange), education (green) and construction (red) metaphors along time.

We further hypothesize systematic differences across interactional roles. Preliminary observations (Figure 3) suggest that metaphor usage is more homogeneous across users when they act as commenters, whereas greater variability emerges when individuals narrate their own illness trajectories. This pattern may indicate stronger alignment pressures in supportive responses and greater idiolectal variation in self-narration.

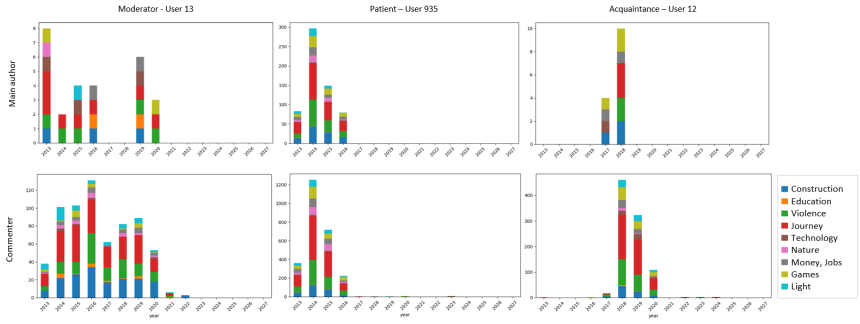

Figure 3. Metaphor usage from particular users when interacting as main author of the post vs commenter.

By conceptualizing metaphor use as a dynamic, socially distributed process, this study proposes an empirically grounded framework for examining conversational mechanisms such as alignment, diffusion, and the emergence of conventional patterns in online health communities. More broadly, the findings may contribute to understanding how micro-level interactional processes scale up to patterns of linguistic evolution.

Acknowledgements

Project PID2024-159530OB-I00 funded by MICIU /AEI /10.13039/501100011033 / FEDER, UE; by DGA Government predoctoral fellowship, and by Proyecto PAPIIT IG400325.

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